Validation of the Risk Lab market-risk models

VaR / ES estimation, backtesting and 10-day scaling on synthetic data

TEXT SLOT 1: *Author, function and date of the validation.*

DRAFT – this version has 19 open text slots (orange boxes). All figures and tables are generated from synthetic data by `report/build_report.py`.

Contents

1 Executive summary **1**

2 Scope and model description **2**

3 Data **2**
3.1 Synthetic design . . . 2
3.2 Data quality: survivorship bias . . . 2

4 Conceptual soundness **3**

5 Implementation verification **4**

6 Outcome analysis and backtesting **5**

7 Benchmarking and sensitivity **7**
7.1 Estimation window . . . 7
7.2 10-day horizon . . . 8

8 Limitations **9**

9 Findings and recommendations **9**

A Formulas and test definitions **10**

B Reproduction **11**

1 Executive summary

TEXT SLOT 2: *Purpose and scope of the validation in two or three sentences; the overall opinion (e.g. fit for use with conditions); the most important findings by ID (Table 7).*

TEXT SLOT 3: *The key quantitative result in one sentence each: the size problem of Kupiec at $T = 250$, the estimation risk of the 250-day GARCH window, and the 10-day scaling result.*

2 Scope and model description

TEXT SLOT 4: *What is validated: one-day VaR / ES models, the backtests and the 10-day methods; what is not (portfolio aggregation, real data, production controls).*

Table 1: Model inventory: the models under validation.

Model	Assumptions	Parameters	Window (days)	Refit
Historical simulation	iid losses; the window is representative	none (order statistic $k = \lceil n\alpha \rceil$)	250	daily
Normal	iid normal losses	μ, σ (sample, ddof 1)	250	daily
Student-t	iid location-scale t	μ, s, $\nu \in [2.05, 200]$ (MLE)	250	50 days
EWMA-t	zero mean, $\sigma^2_{t+1} = \lambda\sigma^2_t + (1-\lambda)L^2_t$	$\lambda = 0.94$, $\nu = 6$ fixed	250	50 days + daily update
GARCH(1,1)-t	unit-variance t innovations	$\mu, \omega, \alpha, \beta, \nu$ (MLE, $\alpha + \beta \leq 0.999$)	250 / 1000	50 days + daily update
Filtered HS	GARCH(1,1) filter, empirical residuals	GARCH by Gaussian QMLE; residual quantile	250 / 1000	50 days + daily update
10-day: $\sqrt{10}$ scaling	iid, zero mean (approximation)	one-day model	500	per origin
10-day: GARCH Monte Carlo	GARCH(1,1)-t dynamics	20 000 simulated paths	500	per origin
10-day: block bootstrap	stationary dependence	automatic block length (Politis–White)	500	per origin

TEXT SLOT 5: *One paragraph per model family: intended use, key assumptions, why it is in scope.*

3 Data

3.1 Synthetic design

TEXT SLOT 6: *Why synthetic data: known truth (true conditional VaR / ES), no licence issues; the data-generating processes (iid normal, iid t5, GARCH(1,1)-normal / -t6, regime switching) and their parameters.*

3.2 Data quality: survivorship bias

TEXT SLOT 7: *Why data quality matters for risk models; what the survivorship study shows (selection bias against attrition, and which base rate answers which question) and how it would carry over to real data (point-in-time constituents, delisted firms).*

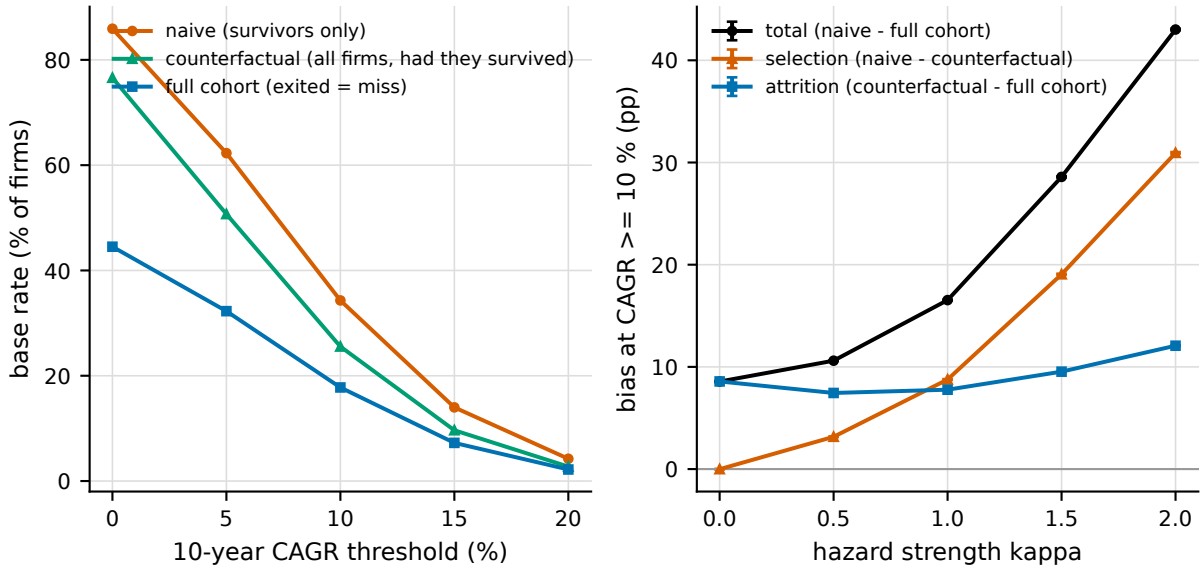


Figure 1: Survivorship bias in 10-year growth base rates, synthetic universe of 2000 firms (M = 2000 cohorts, base hazard h0 = 0.04). Left: the naive, counterfactual and full-cohort base rates for kappa = 1. Right: at a CAGR threshold of 10 %, the total bias and its parts against the hazard strength: selection (survivors are not a random sample) and attrition (exited firms count as misses). Error bars ±1.96 se (Monte-Carlo noise only, mostly smaller than the markers); the size depends on the assumed hazard.

Table 2: Survivorship bias of 10-year growth base rates (in %; synthetic universe of 2000 firms, M = 2000 cohorts per setting). Naive: survivors only ("what share of today's listed firms grew this fast?"). Counterfactual: all firms had they survived ("what share has the growth driver?"). Full cohort: exited firms count as misses ("how likely is a firm picked at the start to achieve it?"). Selection = naive − counterfactual; attrition = counterfactual − full cohort; total = their sum (percentage points ± Monte-Carlo se; the parameter choices dominate the size). Acquisitions are treated like failures.

h_0	κ	Surv. (%)	CAGR ≥ (%)	Naive	Counterf.	Full cohort	Selection	Attrition	Total
0	0	100.0	5	50.7	50.7	50.7	0.0 ± 0.00	0.0 ± 0.00	0.0 ± 0.00
0	0	100.0	10	25.6	25.6	25.6	0.0 ± 0.00	0.0 ± 0.00	0.0 ± 0.00
0	0	100.0	15	9.6	9.6	9.6	0.0 ± 0.00	0.0 ± 0.00	0.0 ± 0.00
0.04	0	66.5	5	50.7	50.7	33.7	-0.0 ± 0.02	17.0 ± 0.02	17.0 ± 0.02
0.04	0	66.5	10	25.6	25.6	17.0	-0.0 ± 0.02	8.6 ± 0.01	8.6 ± 0.01
0.04	0	66.5	15	9.7	9.7	6.4	-0.0 ± 0.01	3.2 ± 0.01	3.2 ± 0.01
0.04	0.5	63.0	5	54.8	50.7	34.5	4.1 ± 0.02	16.2 ± 0.02	20.2 ± 0.02
0.04	0.5	63.0	10	28.7	25.5	18.1	3.2 ± 0.02	7.4 ± 0.01	10.6 ± 0.01
0.04	0.5	63.0	15	11.3	9.6	7.1	1.6 ± 0.01	2.5 ± 0.01	4.2 ± 0.01
0.04	1	51.8	5	62.3	50.7	32.3	11.6 ± 0.02	18.4 ± 0.02	30.0 ± 0.02
0.04	1	51.8	10	34.3	25.5	17.8	8.8 ± 0.02	7.8 ± 0.01	16.5 ± 0.02
0.04	1	51.8	15	14.0	9.6	7.2	4.3 ± 0.01	2.4 ± 0.01	6.7 ± 0.01
0.04	1.5	35.9	5	74.4	50.7	26.7	23.7 ± 0.03	24.0 ± 0.02	47.7 ± 0.03
0.04	1.5	35.9	10	44.6	25.6	16.0	19.1 ± 0.03	9.5 ± 0.01	28.6 ± 0.03
0.04	1.5	35.9	15	19.3	9.6	6.9	9.7 ± 0.02	2.7 ± 0.01	12.4 ± 0.02
0.04	2	23.8	5	85.0	50.6	20.2	34.3 ± 0.04	30.4 ± 0.02	64.7 ± 0.03
0.04	2	23.8	10	56.4	25.5	13.4	30.9 ± 0.04	12.1 ± 0.02	43.0 ± 0.04
0.04	2	23.8	15	26.5	9.6	6.3	16.9 ± 0.04	3.3 ± 0.01	20.2 ± 0.03

4 Conceptual soundness

TEXT SLOT 8: *Assessment of the modelling choices: loss sign convention, quantile convention of historical simulation, distributional assumptions, GARCH estimation (MLE, constraints, starts), the ES definition.*

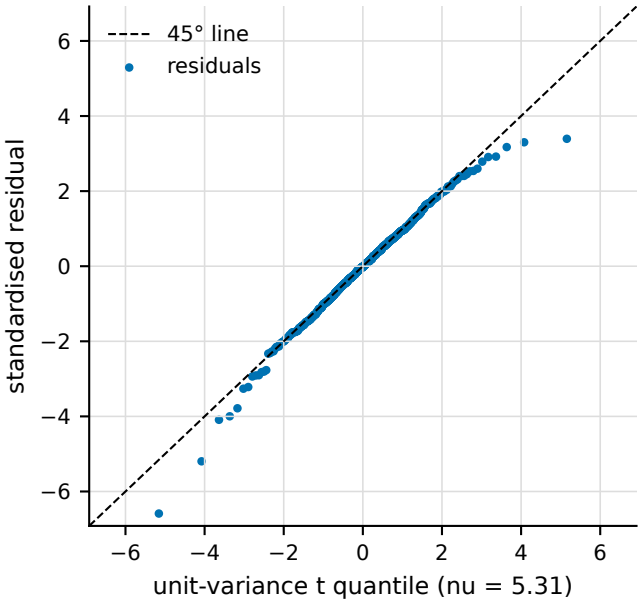


Figure 2: QQ plot of the standardised residuals of a GARCH(1,1)-t fit to one synthetic GARCH-t6 path (seed 20261110, n = 1000) against the fitted unit-variance t ($\hat{\nu} = 5.31$); plotting positions $(i - 0.5)/n$.

TEXT SLOT 9: *Interpretation of the QQ plot: are the fitted t tails adequate?*

5 Implementation verification

TEXT SLOT 10: *Testing approach: unit tests with independent checks (closed forms against numerical integration, brute-force binomial sums, Monte-Carlo checks), look-ahead spy tests, reproducibility (fixed seeds; byte-identical random streams within one numpy version, results files fingerprinted by the code that produced them), and the independent code reviews; the number of tests and where they run.*

6 Outcome analysis and backtesting

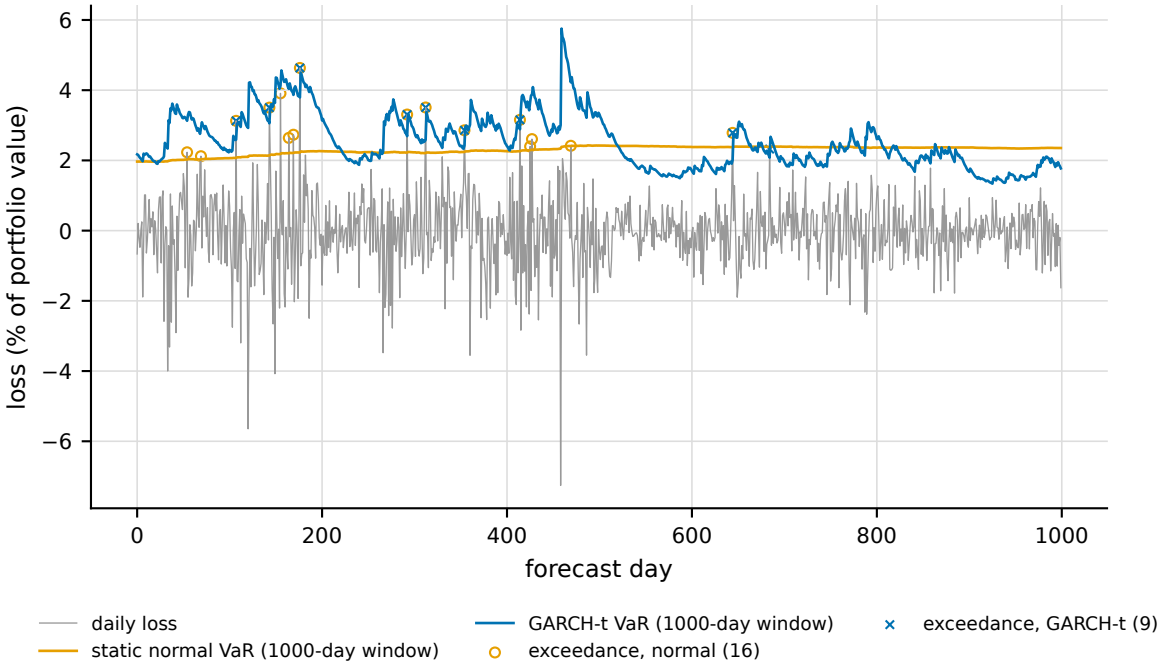


Figure 3: One synthetic GARCH(1,1)-t6 path (seed 20261109, 1000 forecast days): daily losses, the 99 % VaR of a static normal model (1000-day window) and of GARCH(1,1)-t (1000-day window), and their exceedances (normal 16, GARCH-t 9; expected 10). Both models use the same estimation window, so only the model differs. One path only: the counts illustrate clustering, they are not a test.

TEXT SLOT 11: *Reading of the exceedance plot: clustering of the static-normal exceedances against the GARCH-t forecast.*

Table 3: Size of the backtests at 5 % (rejection rate of a correct VaR / ES, in %). Exact binomial sums; oracle rows: M = 1000 synthetic replications (± se).

	Kupiec	Binomial exact	Christoffersen cc	ES Z2
exact size, T = 250	9.48	4.12	–	–
exact size, T = 1000	5.51	3.65	–	–
oracle, garch_normal, T = 250	10.6 ± 1.0	4.5 ± 0.7	1.0 ± 0.3	4.6 ± 0.7
oracle, garch_normal, T = 1000	6.2 ± 0.8	4.6 ± 0.7	2.8 ± 0.5	4.6 ± 0.7
oracle, garch_t6, T = 250	9.6 ± 0.9	3.8 ± 0.6	1.0 ± 0.3	4.4 ± 0.6
oracle, garch_t6, T = 1000	4.6 ± 0.7	2.5 ± 0.5	1.7 ± 0.4	4.6 ± 0.7
oracle, iid_normal, T = 250	8.7 ± 0.9	5.3 ± 0.7	1.1 ± 0.3	6.4 ± 0.8
oracle, iid_normal, T = 1000	5.7 ± 0.7	4.1 ± 0.6	2.7 ± 0.5	4.8 ± 0.7
oracle, iid_t5, T = 250	9.0 ± 0.9	4.0 ± 0.6	1.0 ± 0.3	5.1 ± 0.7
oracle, iid_t5, T = 1000	7.4 ± 0.8	5.0 ± 0.7	4.1 ± 0.6	6.5 ± 0.8
oracle, regime, T = 250	10.5 ± 1.0	4.3 ± 0.6	1.0 ± 0.3	5.0 ± 0.7
oracle, regime, T = 1000	5.1 ± 0.7	3.4 ± 0.6	2.6 ± 0.5	4.4 ± 0.6

TEXT SLOT 12: *Size of the tests: why Kupiec over-rejects at T = 250 and what that means for a one-year backtest.*

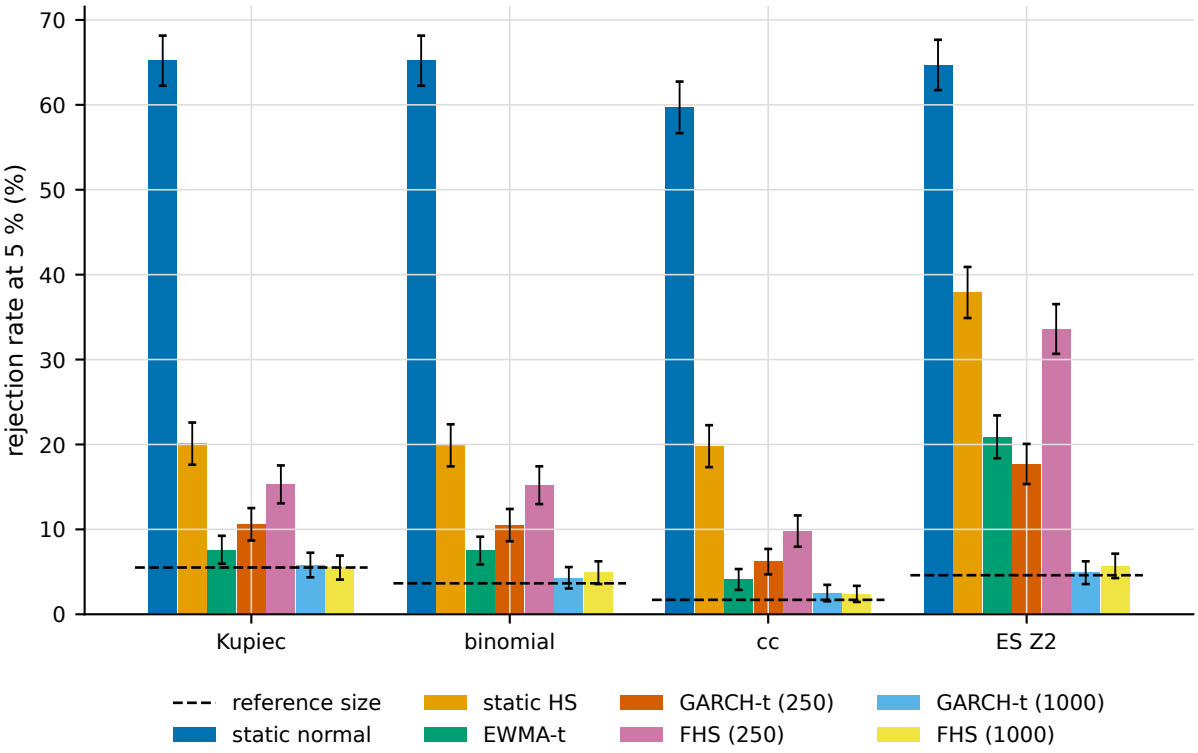


Figure 4: Rejection rates of the VaR tests (Kupiec, exact binomial, Christoffersen cc; 99 %) and the Acerbi–Székely Z2 ES test (97.5 %) on synthetic GARCH(1,1)-t6 data, T = 1000, M = 1000 replications; error bars ±1.96 se. Dashed lines: the reference size of each test (exact size for Kupiec and binomial, oracle rate for cc and Z2). Numbers in brackets: estimation window in days.

Table 4: Rejection rates (%, ± se) of the backtests at 5 % by data, model and estimation window; M = 1000 synthetic replications. Not size-adjusted: compare with Table 3.

Data	Model	Window	T	Exceed. (%)	Kupiec	Binomial	cc	Z2
iid_t5	normal	250	250	1.58	10.8 ± 1.0	19.2 ± 1.2	5.4 ± 0.7	18.2 ± 1.2
garch_t6	normal	250	250	1.85	27.5 ± 1.4	32.4 ± 1.5	20.8 ± 1.3	32.2 ± 1.5
garch_t6	hs	250	250	1.35	19.3 ± 1.2	17.4 ± 1.2	9.8 ± 0.9	24.5 ± 1.4
garch_t6	ewma_t	250	250	1.28	6.3 ± 0.8	9.1 ± 0.9	1.9 ± 0.4	11.1 ± 1.0
garch_t6	garch_t	250	250	1.25	12.0 ± 1.0	11.7 ± 1.0	4.0 ± 0.6	12.2 ± 1.0
garch_t6	fhs	250	250	1.35	11.1 ± 1.0	14.1 ± 1.1	4.9 ± 0.7	16.9 ± 1.2
regime	normal	250	250	2.27	37.8 ± 1.5	48.2 ± 1.6	37.1 ± 1.5	43.5 ± 1.6
regime	hs	250	250	1.35	14.6 ± 1.1	16.5 ± 1.2	10.3 ± 1.0	25.0 ± 1.4
regime	ewma_t	250	250	1.33	6.3 ± 0.8	10.5 ± 1.0	4.3 ± 0.6	13.9 ± 1.1
regime	garch_t	250	250	1.24	11.6 ± 1.0	11.1 ± 1.0	4.9 ± 0.7	12.2 ± 1.0
regime	fhs	250	250	1.36	10.9 ± 1.0	14.2 ± 1.1	5.7 ± 0.7	17.5 ± 1.2
iid_t5	normal	250	1000	1.58	40.6 ± 1.6	40.6 ± 1.6	24.5 ± 1.4	35.6 ± 1.5
garch_t6	normal	250	1000	1.87	65.2 ± 1.5	65.2 ± 1.5	59.7 ± 1.6	64.7 ± 1.5
garch_t6	hs	250	1000	1.38	20.1 ± 1.3	19.9 ± 1.3	19.8 ± 1.3	37.9 ± 1.5
garch_t6	ewma_t	250	1000	1.24	7.6 ± 0.8	7.5 ± 0.8	4.1 ± 0.6	20.9 ± 1.3
garch_t6	garch_t	250	1000	1.25	10.6 ± 1.0	10.5 ± 1.0	6.2 ± 0.8	17.7 ± 1.2
garch_t6	fhs	250	1000	1.35	15.3 ± 1.1	15.2 ± 1.1	9.8 ± 0.9	33.6 ± 1.5
garch_t6	garch_t	1000	1000	1.03	5.8 ± 0.7	4.3 ± 0.6	2.5 ± 0.5	4.9 ± 0.7
garch_t6	fhs	1000	1000	1.12	5.5 ± 0.7	4.9 ± 0.7	2.4 ± 0.5	5.7 ± 0.7
regime	normal	250	1000	2.28	92.4 ± 0.8	92.4 ± 0.8	89.6 ± 1.0	86.8 ± 1.1
regime	hs	250	1000	1.35	16.1 ± 1.2	16.1 ± 1.2	21.6 ± 1.3	38.8 ± 1.5
regime	ewma_t	250	1000	1.37	19.4 ± 1.3	19.1 ± 1.2	19.2 ± 1.2	35.1 ± 1.5
regime	garch_t	250	1000	1.27	12.8 ± 1.1	12.4 ± 1.0	9.5 ± 0.9	18.0 ± 1.2
regime	fhs	250	1000	1.37	17.4 ± 1.2	17.4 ± 1.2	12.2 ± 1.0	34.7 ± 1.5

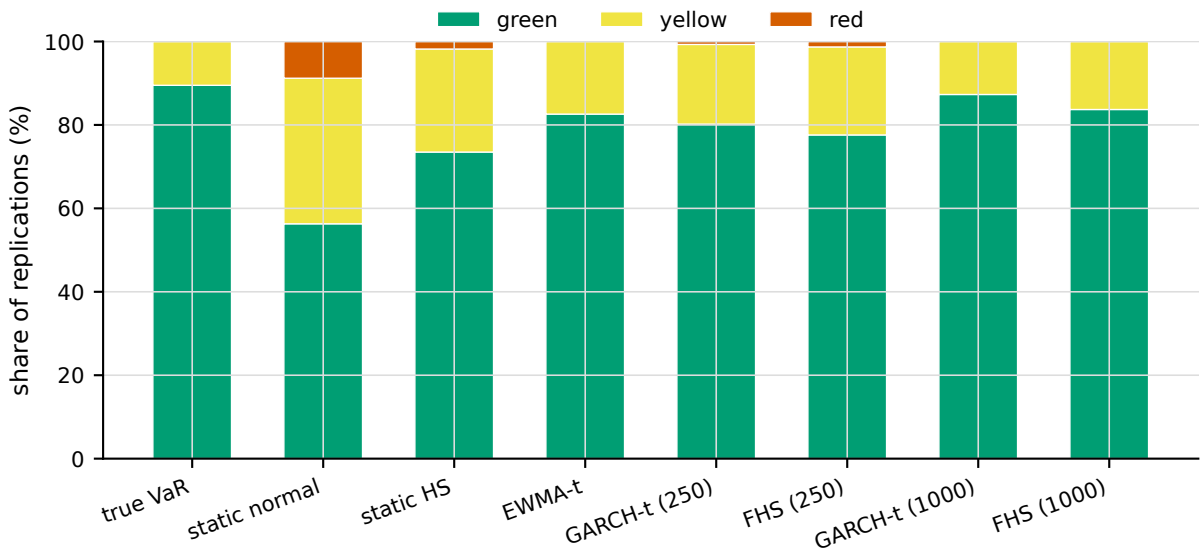


Figure 5: Basel traffic-light zones (last 250 forecast days, 99 % VaR) per model on synthetic GARCH(1,1)-t6 data, T = 1000, M = 1000. Numbers in brackets: estimation window in days.

TEXT SLOT 13: *Power and the comparison of static and conditional models; the ES test against the VaR tests; the traffic light as the regulatory view.*

7 Benchmarking and sensitivity

7.1 Estimation window

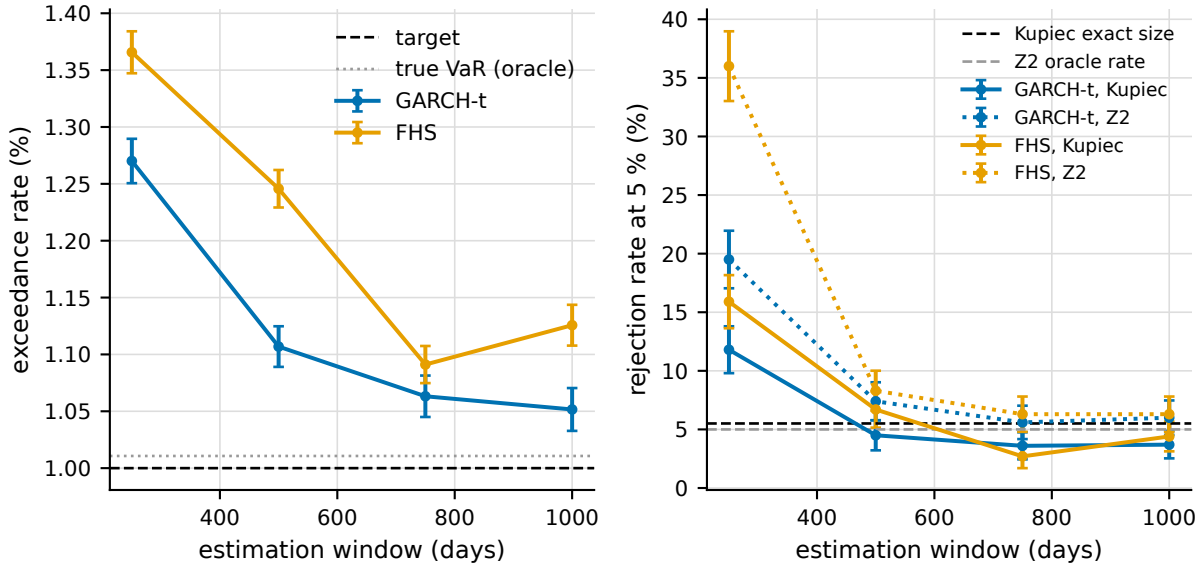


Figure 6: Estimation risk against the window length: correctly specified GARCH(1,1)-t and FHS on synthetic GARCH-t6 data, T = 1000 (the same forecast days for every window), M = 1000. Left: exceedance rate of the 99 % VaR, with the target and the oracle rate (the left axis does not start at 0); right: Kupiec and Z2 rejection rates with their reference sizes. Error bars ±1.96 se.

Table 5: Estimation window: correctly specified GARCH(1,1)-t and FHS on synthetic GARCH-t6 data, T = 1000 (same forecast days for every window), M = 1000; rates in % (± se). Reference sizes: Kupiec 5.51, binomial 3.65 (exact); cc and Z2: the true-VaR row. Δ: paired difference of the exceedance rate against the true VaR on the same days (percentage points ± se); positive = anti-conservative. The 1000-day rows use the same design as the 1000-day rows of the power study (checked per replication); only the data seeds differ, so differences between the two studies are Monte-Carlo noise.

Model	Window	Exceed. (%)	Δ exceed. (pp)	Kupiec	Binomial	cc	Z2	VaR error (%)
true VaR	–	1.01 ± 0.01	–	4.7 ± 0.7	3.4 ± 0.6	2.0 ± 0.4	5.0 ± 0.7	0.0
garch_t	250	1.27 ± 0.01	+0.259 ± 0.010	11.8 ± 1.0	11.6 ± 1.0	6.7 ± 0.8	19.5 ± 1.3	11.6
garch_t	500	1.11 ± 0.01	+0.096 ± 0.008	4.5 ± 0.7	4.2 ± 0.6	2.4 ± 0.5	7.4 ± 0.8	7.5
garch_t	750	1.06 ± 0.01	+0.052 ± 0.007	3.6 ± 0.6	2.7 ± 0.5	2.2 ± 0.5	5.6 ± 0.7	6.0
garch_t	1000	1.05 ± 0.01	+0.041 ± 0.007	3.7 ± 0.6	2.9 ± 0.5	2.4 ± 0.5	6.0 ± 0.8	5.1
fhs	250	1.37 ± 0.01	+0.355 ± 0.013	15.9 ± 1.2	15.9 ± 1.2	9.8 ± 0.9	36.0 ± 1.5	15.5
fhs	500	1.25 ± 0.01	+0.235 ± 0.011	6.7 ± 0.8	6.7 ± 0.8	5.1 ± 0.7	8.3 ± 0.9	10.5
fhs	750	1.09 ± 0.01	+0.080 ± 0.010	2.7 ± 0.5	2.4 ± 0.5	1.6 ± 0.4	6.3 ± 0.8	8.7
fhs	1000	1.13 ± 0.01	+0.115 ± 0.010	4.4 ± 0.6	4.3 ± 0.6	2.9 ± 0.5	6.3 ± 0.8	7.3

TEXT SLOT 14: *How the over-rejection of the correct model falls with the window length, why the VaR stays slightly anti-conservative even where the rejection rates are back at the reference size (paired exceedance difference), and the recommended window.*

7.2 10-day horizon

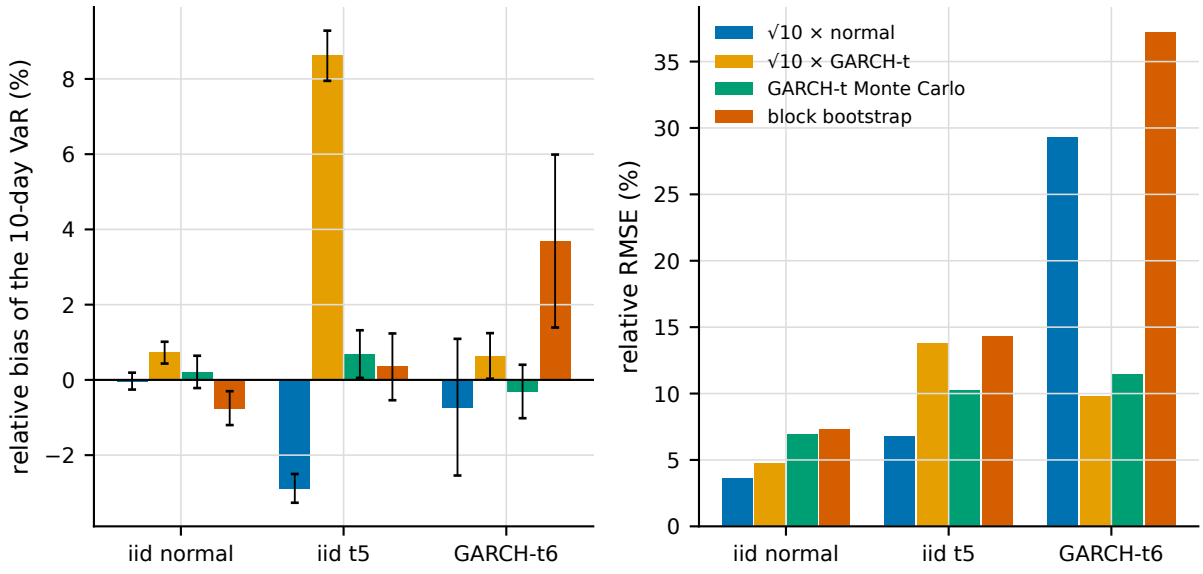


Figure 7: 10-day 99 % VaR against the true 10-day VaR: relative bias (±1.96 se) and relative RMSE per method and synthetic data set; 500-day window, M = 1000.

Table 6: 10-day 99 % VaR against the true 10-day VaR (relative, in %; ± se over M = 1000 replications). MC se: simulation error of the estimate and of the truth.

Data	Method	Bias	RMSE	Est. MC se	Truth MC se
iid_normal	sqrt10_normal	-0.0 ± 0.1	3.6	–	0.00
iid_normal	sqrt10_garch	+0.7 ± 0.1	4.7	–	0.00
iid_normal	garch_mc	+0.2 ± 0.2	6.9	1.18	0.00
iid_normal	bootstrap	-0.8 ± 0.2	7.3	1.15	0.00
iid_t5	sqrt10_normal	-2.9 ± 0.2	6.8	–	0.84
iid_t5	sqrt10_garch	+8.6 ± 0.3	13.8	–	0.84
iid_t5	garch_mc	+0.7 ± 0.3	10.2	1.41	0.84
iid_t5	bootstrap	+0.3 ± 0.5	14.3	1.31	0.84
garch_t6	sqrt10_normal	-0.7 ± 0.9	29.3	–	1.01
garch_t6	sqrt10_garch	+0.6 ± 0.3	9.8	–	1.01
garch_t6	garch_mc	-0.3 ± 0.4	11.5	1.60	1.01
garch_t6	bootstrap	+3.7 ± 1.2	37.2	1.44	1.01

TEXT SLOT 15: *Square-root-of-time against simulation and bootstrap: when the approximation is acceptable.*

8 Limitations

TEXT SLOT 16: *Synthetic data only; univariate; fixed parameters of the data-generating processes; Monte-Carlo error of the study results (see the ± se); what a validation on real data would add.*

9 Findings and recommendations

TEXT SLOT 17: *How the severities were set (definitions of High / Medium / Low) and the overall rating.*

Table 7: Findings register. Severities are PROPOSED and must be confirmed by the validator.

ID	Finding	Severity	Evidence	Proposed remediation
F1	With a 250-day window the correctly specified GARCH(1,1)-t over-rejects: Kupiec 10.6 ± 1.0 % vs exact size 5.5 %, Z2 17.7 ± 1.2 % (T = 1000); exceedance rate 1.25 %. With 1000 days: Kupiec 5.8 ± 0.7 %. Estimation-window study: the rejection rates fall to about the reference size by 500 days (GARCH-t) and 750 days (FHS) (Kupiec, % by window: 250: 11.8, 500: 4.5, 750: 3.6, 1000: 3.7), but the VaR stays slightly anti-conservative: exceedance rate 1.05–1.13 % against the true VaR's 1.01 % (GARCH-t by window: 250: 1.27, 500: 1.11, 750: 1.06, 1000: 1.05). Paired difference against the true VaR at 1000 days: +0.041 ± 0.007 pp.	PROPOSED: High	Tab. 4, 5; Fig. 6	Estimation window of at least 500 days for GARCH-t and 750 days for FHS (where the rejection rates reach the reference sizes); account for the remaining small anti-conservatism (parameter uncertainty), e.g. by a VaR add-on or a longer window.
F2	FHS with a 250-day window underestimates the tail (ES): Z2 rejects in 33.6 ± 1.5 % (oracle 4.6 %, T = 1000); with 750 days 6.3 ± 0.8 %.	PROPOSED: Medium	Tab. 4, 5	Longer window for the residual distribution; monitor the ES backtest separately from VaR.
F3	Static models miss volatility clustering: the static normal VaR on GARCH-t data is rejected by cc in 59.7 ± 1.6 % and by Z2 in 64.7 ± 1.5 % (T = 1000).	PROPOSED: High	Fig. 3, 4; Tab. 4	Use a conditional volatility model (GARCH-t / FHS with an adequate window).
F4	The Kupiec test is mis-sized at T = 250: exact size 9.48 % at a nominal 5 % (x = 0 is rejected); the exact binomial test keeps 4.12 %.	PROPOSED: Medium	Tab. 3	Report the exact binomial test (or the exact size) alongside Kupiec for one-year backtests.
F5	The Christoffersen independence test is undersized (0.9–2.6 % on correct VaRs) and has little power on its own.	PROPOSED: Low	Results file of the power study (ind columns)	Do not rely on the independence test alone; use cc and the ES test.
F6	$\sqrt{10}$ scaling of a fat-tailed one-day VaR overstates the 10-day VaR on iid t5 data (+8.6 ± 0.3 %). On GARCH-t6 data $\sqrt{10}$ is unbiased on average but $\sqrt{10}\times$ static normal has an RMSE of 29.3 %.	PROPOSED: Medium	Fig. 7; Tab. 6	Simulate the 10-day distribution (GARCH Monte Carlo); use $\sqrt{10}$ only for a conditional one-day VaR when the 10-day sum is near-normal.
F7	The stationary block bootstrap is biased on GARCH-t6 data (+3.7 ± 1.2 %, RMSE 37.2 %): it does not condition on the current volatility.	PROPOSED: Low	Tab. 6	Do not use the unconditional bootstrap as the 10-day benchmark under volatility clustering.
F8	Historical simulation with $k = \lceil n\alpha \rceil$ has a finite-sample exceedance probability of $(n-k+1)/(n+1) = 1.20\,\%$ at n = 250, $\alpha = 0.99$ (simulated: 1.21 % on iid normal data).	PROPOSED: Low	Results of the power study (HS on iid normal data)	Document the quantile convention; consider a longer window or interpolation.
F9	Survivorship (synthetic, hazard strength 1, CAGR ≥ 10 %): a survivors-only database shows 34.3 %, against 25.5 % for all firms had they survived (selection bias 8.8 ± 0.02 pp) and 17.8 % for the full cohort with exits as misses (attrition 7.8 ± 0.01 pp). The size depends on the assumed hazard; the ± se is Monte-Carlo noise only.	PROPOSED: Medium	Fig. 1; Tab. 2	Use point-in-time data that include delisted firms, and state which base rate (counterfactual or full cohort) answers the question asked.

TEXT SLOT 18: *Recommendations and follow-ups, with owners and dates.*

A Formulas and test definitions

Losses are positive: $L = -r$. For a confidence level α, $\mathrm{VaR}_\alpha = F_L^{-1}(\alpha)$ and $\mathrm{ES}_\alpha = \mathrm{E}[L \mid L \geq \mathrm{VaR}_\alpha]$.

Kupiec (proportion of failures). With T days, x exceedances, $p = 1 - \alpha$, $\hat{p} = x/T$:

$$LR_{uc} = -2 \ln \frac{(1-p)^{T-x} p^x}{(1-\hat{p})^{T-x} \hat{p}^x} \sim \chi^2_1.$$

Exact binomial test. Two-sided p-value: the total probability under $\mathrm{Bin}(T, p)$ of all counts at most as likely as x.

Christoffersen. With transition counts n_{ij}, $LR_{ind} = -2 \ln[L(\hat{\pi})/L(\hat{\pi}_{01}, \hat{\pi}_{11})] \sim \chi^2_1$ and $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2_2$, both on the $T - 1$ transitions.

Acerbi–Székely Z2 (loss sign).

$$Z_2 = \sum_{t=1}^{T} \frac{L_t I_t}{T(1-\alpha)\,\mathrm{ES}_t} - 1, \qquad I_t = \mathbf{1}\{L_t > \mathrm{VaR}_t\};$$

$Z_2 > 0$ indicates underestimated ES; the p-value is simulated under each day's predictive distribution.

GARCH(1,1). $L_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(L_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$, z_t unit-variance t_ν.

Basel traffic light. Zones from the binomial CDF of the exceedances in 250 days: green below 95 %, red from 99.99 %.

TEXT SLOT 19: *Any further derivations the author wants to include (e.g. the normal and t ES formulas).*

B Reproduction

From `UFOS.ai/RiskLab/`:

```
python -m unittest discover -s tests
python studies/var_backtest_power.py --mode full --save-results
python studies/horizon_10d.py --mode full --save-results
python studies/estimation_window.py --mode full --save-results
python studies/survivorship.py --mode full --save-results
python report/build_report.py --publish
```

Each study command also takes the option `--save-results`, which writes its full-run results to the folder `studies/results/`. The study results used here are stored there, each with a fingerprint of the code that produced it; the build refuses results whose fingerprint does not match the current code.